Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A condenser of capacity C is charged to a potential difference of V_1. The plates of the condenser are then connected to an ideal inductor of inductance L. The current through the inductor when the potential difference across the condenser reduces to V_2 is

(A) $\left(\frac{C\left(V_1^2-V_2^2\right)}{L}\right)^{1/2}$

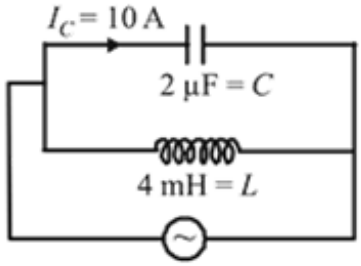
(B) $\left(\frac{C(V_1-V_2)^2}{L}\right)^{1/2}$

(C) $\frac{C\left(V_1^2-V_2^2\right)}{L}$

(D) $\frac{C(V_1-V_2)}{L}$

Q.2

For the circuit as shown in figure; the applied current in A.C. circuit is zero ampere and $I_C = 10\ \mathrm{A}$. Then the magnitude of current I_L is



(A) $4\ \mathrm{A}$

(B) $10\ \mathrm{A}$

(C) $5\ \mathrm{A}$

(D) undefined

Q.3

What is the value of inductance L for which the current is maximum in a series LCR circuit with $\mathrm{C} = 10\mu\mathrm{F}$ and $\omega = 1000\ \mathrm{s}^{-1}$?

(A) $1\mathrm{mH}$

(B) cannot be calculated unless R is known

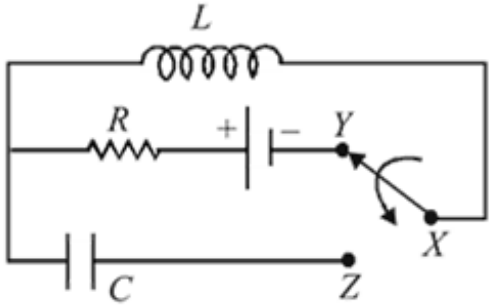
(C) $10\mathrm{mH}$

(D) $100\mathrm{mH}$

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Q.4

In the circuit shown, the symbols have their usual meanings. The cell has emf E. X is initially joined to Y for a long time. Then, X is joined to Z. The maximum charge on C at any later time will be



(A) $\frac{E}{R\sqrt{LC}}$

(B) $\frac{ER}{2\sqrt{LC}}$

(C) $\frac{E\sqrt{LC}}{2R}$

(D) $\frac{E\sqrt{LC}}{R}$

Q.5

A coil of 40 henry inductance is connected in series with a resistance of 8 ohm and the combination is joined to the terminals of a 2 volt battery. The time constant of the circuit is

(A) 20 seconds

(B) 5 seconds

(C) $1/5$ seconds

(D) 40 seconds

Q.6

A resistor and an inductor are connected to an ac supply of $120\ \mathrm{V}$ and $50\ \mathrm{Hz}$. The current in the circuit is $3\ \mathrm{A}$. If the power consumed in the circuit is $108\ \mathrm{W}$, then the resistance in the circuit is

(A) 12Ω

(B) 40Ω

(C) $\sqrt{(52 \times 25)}\Omega$

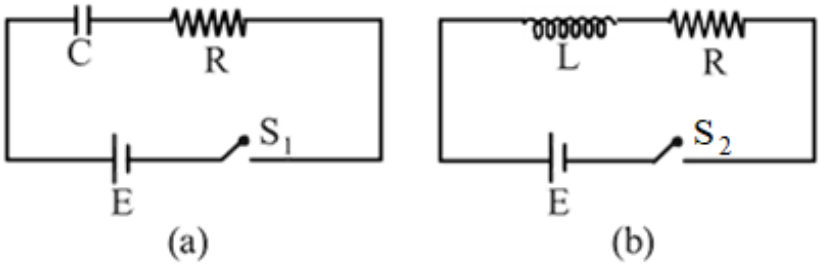
(D) 360Ω

Q.7

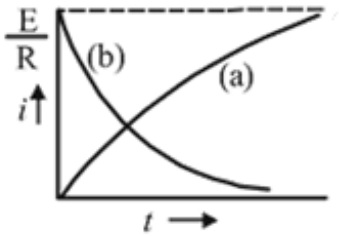
In the circuits (a) and (b) switches S_1 and S_2 are closed at $\mathrm{t} = 0$ and are kept closed for a long time. The variation of current in the two circuits for $\mathrm{t} \geq 0$ are roughly shown by figure (figures are schematic and not drawn

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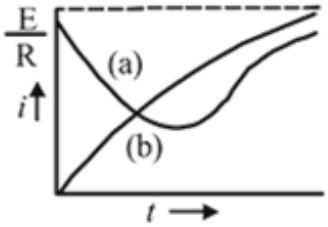
to scale):



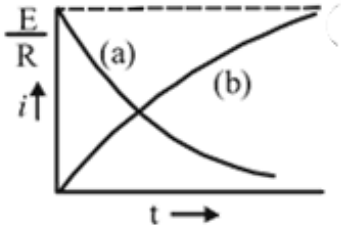
(A)



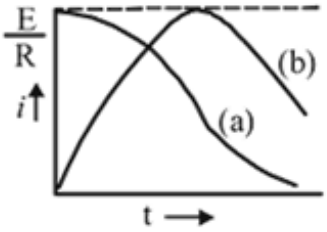
(B)



(C)

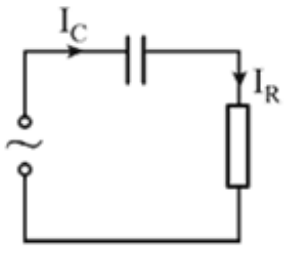


(D)

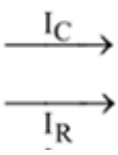


Q.8

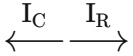
Figure shows a source of alternating voltage connected to a capacitor and resistor. Which of the following phasor diagrams correctly describes the phase relationship between I_c, the current between the source and the capacitor, and I_R, the current in the resistor?

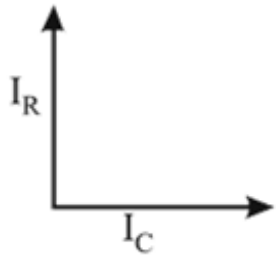


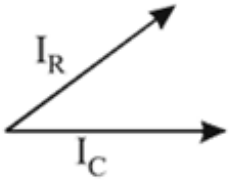
(A)



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(B) 

(C) 

(D) 

Q.9

A capacitor of $10\mu\text{F}$ and an inductor of 1H are joined in series. An ac source of $50\ \text{Hz}$ is applied to this combination. What is the impedance of the combination?

(A) $\frac{5\left(\pi^2-5\right)}{\pi}\Omega$

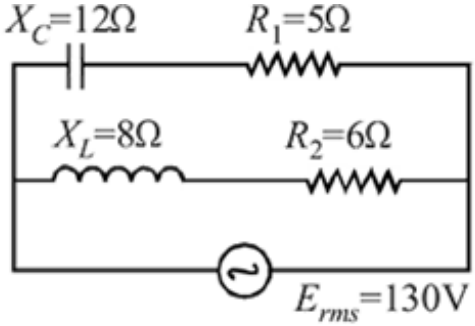
(B) $\frac{10^2\left(10-\pi^2\right)}{\pi}\Omega$

(C) $\frac{10\left(\pi^2-5\right)}{\pi}\Omega$

(D) $\frac{5^2\left(10-\pi^2\right)}{\pi}\Omega$

Q.10

What is the amount of power delivered by the ac source in the circuit shown (in watts).



(A) 500 watt

(B) 1014 watt

(C) 1514 watt

(D) 2013 watt

Q.11

The current in an $L-R$ circuit builds up to $(3/4)^{\text{th}}$ of its steady state value in 4 seconds. The time constant of this circuit is

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(A) $\frac{1}{\ln 2}\text{sec}$

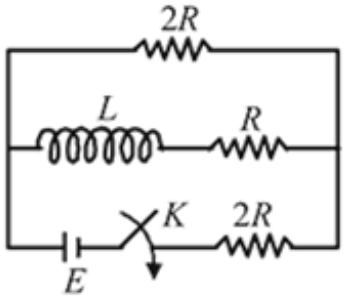
(B) $\frac{2}{\ln 2}\text{sec}$

(C) $\frac{3}{\ln 2}\text{sec}$

(D) $\frac{4}{\ln 2}\text{sec}$

Q.12

Time constant of $L - R$ circuit will be



(A) $\frac{L}{R}$

(B) $\frac{2L}{R}$

(C) $\frac{L}{2R}$

(D) None of these

Q.13

LC circuit contains a 20mH inductor and a $50\mu\text{F}$ capacitor with an initial charge of 10mC. The resistance of the circuit is negligible. Let the instant the circuit is closed at $\text{t} = 0$. At what time is the energy stored completely magnetic?

(A) $\text{t} = 0$

(B) $\text{t} = 1.57\ \text{ms}$

(C) $\text{t} = 3.14\ \text{ms}$

(D) $\text{t} = 6.28\ \text{ms}$

Q.14

An inductor 20×10^{-3} Henry, a capacitor $100\mu\text{F}$ and a resistor 50Ω are connected in series across a source of $\text{EMF V} = 10\sin 314\text{t}$. If resistance is removed from the circuit and the value of inductance is doubled, then the variation of current with time in the new circuit is

(A) $0.52\cos 314t$

(B) $0.52\sin 314t$

(C) $0.52\sin(314\text{t} + \pi/3)$

(D) None of these

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Q.15

In a LCR circuit capacitance is changed from C to $2C$. For the resonant frequency to remain unchanged, the inductance should be change from L to

(A) $4L$

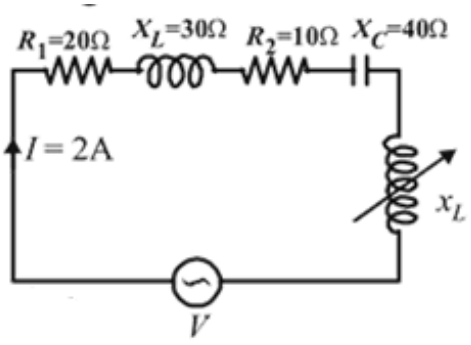
(B) $2L$

(C) $L/2$

(D) $L/4$

Q.16

As shown in figure, value of inductive reactance X_L will be if source voltage is 100 volt



(A) 40Ω

(B) 30Ω

(C) 50Ω

(D) can have any value

Q.17

In a series $\mathrm{L-C-R}$ circuit, $\mathrm{C}=10^{-11}$ Farad, $\mathrm{L}=10^{-5}$ Henry and $\mathrm{R}=100\ \mathrm{Ohm}$, when a constant $\mathrm{D.C.}$ voltage E is applied to the circuit, the capacitor acquires a charge $10^{-9}\mathrm{C}$. The D.C. source is replaced by a sinusoidal voltage source in which the peak voltage E_0 is equal to the constant $\mathrm{D.C.}$ voltage E. At resonance the peak value of the charge acquired by the capacitor will be:

(A) $10^{-15}\mathrm{C}$

(B) $10^{-6}\mathrm{C}$

(C) $10^{-10}\mathrm{C}$

(D) $10^{-8}\mathrm{C}$

Q.18

A capacitor of capacitance $1\mu\mathrm{F}$ is charged to a potential of $1\ \mathrm{V}$. It is connected in parallel to an inductor of inductance $10^{-3}\mathrm{H}$. The maximum current that will flow in the circuit has the value

(A) $\sqrt{1000}\ \mathrm{mA}$

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(B) $1\ \mathrm{mA}$

(C) $1\mu\mathrm{A}$

(D) $1000\ \mathrm{mA}$

Q.19

An ideal coil of inductance 10 henry is joined in series with a resistance of $5\ \mathrm{ohm}$ and a battery of 5 volt. Two seconds after joining, the current flowing in ampere in the circuit will be

(A) e^{-1}

(B) $\left(1-e^{-1}\right)$

(C) $(1-e)$

(D) e

Q.20

A $100\ \ \mathrm{V}$ AC source of frequency $500\ \mathrm{Hz}$ is connected to a LCR circuit with $\mathrm{L}=8.1\ \ \mathrm{mH}$, $\mathrm{C}=12\ .5\ \ \mu\mathrm{F}$ and $\mathrm{R}=10\ \Omega$, all connected in series. The potential difference across the resistance is

(A) $100\ \mathrm{V}$

(B) $200\ \mathrm{V}$

(C) $300\ \mathrm{V}$

(D) $400\ \mathrm{V}$

Answers & Solutions

Q.1 Answer:

$\left(\frac{C\left(V_1^2-V_2^2\right)}{L}\right)^{1/2}$

Solution:

$q=CV_1\cos\omega t \Rightarrow i=\frac{dq}{dt}=-\omega Cv_1\sin\omega t$

Also, $\omega^2=\frac{1}{LC}$ **and** $V=V_1\cos\omega t$

At $t=t_1, V=V_2$ **and** $i=-\omega CV_1\sin\omega t_1$

$\therefore \cos\omega t_1=\frac{V_2}{V_1}$ **(-ve sign gives direction)**

Hence, $i=V_1\sqrt{\frac{C}{L}}\left(1-\frac{V_2^2}{V_1^2}\right)^{1/2}=\left(\frac{C\left(V_1^2-V_2^2\right)}{L}\right)^{1/2}$

Q.2 Answer:

$10\ \mathrm{A}$

Solution:

$I=I_C+(-I_L)$

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Because I_C and I_L are 180° out of phase with each other.

Q.3 Answer:

$100\mathrm{mH}$

Solution:

Condition for which the current is maximum in a series LCR circuit is,

$$\omega = \frac{1}{\sqrt{LC}}$$

$$1000 = \frac{1}{\sqrt{\mathrm{L}\left(10 \times 10^{-6}\right)}} \Rightarrow L = 100\mathrm{mH}$$

Q.4 Answer:

$\frac{E\sqrt{LC}}{R}$

Solution:

Current in inductor $= \frac{E}{R}$

$\therefore$ its energy $= \frac{1}{2}\frac{LE^2}{R^2}$

Same energy is later stored in capacitor

$\frac{Q^2}{2C} = \frac{1}{2}\frac{LE^2}{2} \Rightarrow Q = \sqrt{LC}\frac{E}{R}$

Q.5 Answer:

5 seconds

Solution:

Time constant is L/R

Given, $L = 40\mathrm{H}$ &$R = 8\Omega$

$\therefore \tau = 40/8 = 5\mathrm{sec}.$

Q.6 Answer:

12Ω

Solution:

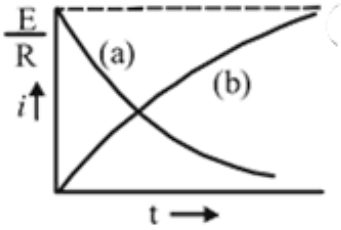
In an ac circuit, a pure indcutor does not consume any power. Therefore, power is consumed by the resistor only.

$\therefore \quad \mathrm{P} = I_v^2 R$

or $108 = (3)^2\mathrm{R}$ **or** $\mathrm{R} = 12\Omega$

Q.7 Answer:

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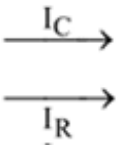
Solution:

For capacitor circuit, $i = i_0 e^{-t/RC}$

For inductor circuit, $i = i_0 \left(1 - e^{-\frac{Rt}{L}}\right)$

Hence graph (c) correctly depicts i **versus** t **graph.**

Q.8 Answer:



Solution:

Since the cpacitor is connected in series to the resistor, current $\mathrm{I_C}$ **from the supply and** $\mathrm{I_R}$ **through the resistor is in phase as represented by choice (a).**

Q.9 Answer:

$\frac{10^2\left(10-\pi^2\right)}{\pi}\Omega$

Solution:

Here, $X_L = \omega_\mathrm{L} = 2\pi f\ \mathrm{L} = 2\pi \times 50 \times 1 = 100\pi\Omega$

$\mathrm{X_c} = \frac{1}{\omega C} = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 50 \times 10 \times 10^{-6}} = \frac{10^3}{\pi}\Omega$

So, X

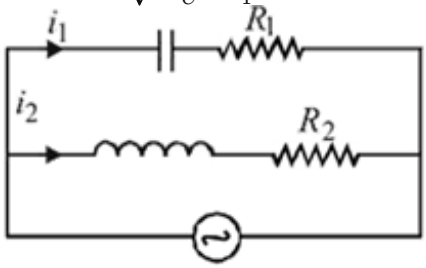
$= |X_L - X_C| = \left|100\pi - \frac{10^3}{\pi}\right| = |\ 10^2\left[\frac{\pi^2-10}{\pi}\right]\ \Omega$

Q.10 Answer:

1514 watt

Solution:

$i_{1\mathrm{rms}} = \frac{E_{\mathrm{rms}}}{\sqrt{X_C^2+R_1^2}} = \frac{130}{13} = 10\ \mathrm{A}$



$i_{2\mathrm{rms}} = \frac{E_{\mathrm{rms}}}{\sqrt{X_L^2+R_2^2}}$

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$= 13$ A

Power dissipated $= i_{1\text{rms}}^2 R_1 + i_{2\text{rms}}^2 R_2 = 10^2 \times 5 + 13^2 + 6$

= power delivered by battery

$= 500 + 169 \times 6 = 1514$ **watt**

Q.11 Answer:

$\frac{2}{\ln 2}\text{sec}$

Solution:

$I = I_0\left(I - e^{-t/\tau}\right)$

where $\tau \rightarrow$ **time constant**

$\therefore \quad \frac{3}{4}I_0 = I_0\left(1 - e^{-t/\tau}\right) \Rightarrow \frac{3}{4} = l - e^{-t/\tau}$

$e^{-t/\tau} = \frac{1}{4}$

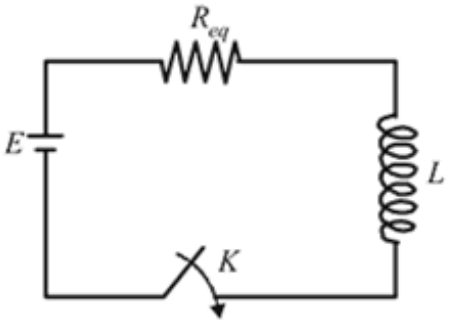
$\frac{-t}{\tau}\ln e = \ln\frac{1}{4} \Rightarrow \frac{-4}{\tau} = -2\ln 2 \Rightarrow \tau = \frac{2}{\ln 2}$

Q.12 Answer:

$\frac{L}{2R}$

Solution:

A circuit equivalent to the given circuit can be drawn as follows:



Where $R_{\text{eq}} = R + \frac{2R \times 2R}{4R}$

$= R + R$

$R_{\text{eq}} = 2R$

$\therefore$ **Time constant** $= \frac{L}{2R}$

Q.13 Answer:

$\text{t} = 1.57$ ms

Solution:

(i) Time period, $\text{T} = 1/\text{f} = 6.28$ s

total charge on capacitor at time $\text{t} = \text{Q}' = \text{Q}\cos\left(\frac{2\pi}{\text{T}}\text{t}\right)$

For energy stored in electrical, we can write, $\text{Q}' = \pm\text{Q}$

It can be inferred that the energy stored in capacitor is completely electrical at time, $\text{t} =$

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$0,\ \mathrm{T}/2, 3\ \mathrm{T}/2, \ldots$

(ii) Magnetic energy is maximum when the electrical energy is zero. hence, $\mathrm{t} = \mathrm{T}/4, 3\ \mathrm{T}/4, 5\ \mathrm{T}/4, \ldots$

Q.14 Answer:

$0.52 \cos 314t$

Solution:

$\mathrm{x_C} - \mathrm{x_L} = 31.85 - 2(6.28) = 19.29$

$I_m = \frac{10}{19.29} = 0.52$

Hence $I = 0.52 \sin(314t + \pi/2) = 0.52 \cos 314$

Q.15 Answer:

$L/2$

Solution:

Resonance frequency,

$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{L'(2C)}} \quad \therefore \quad L' = \frac{L}{2}$

Q.16 Answer:

50Ω

Solution:

$V = I\sqrt{(20+10)^2 + (X_L + X_L - X_C)^2}$

$100 = 2\sqrt{(30)^2 + (30 + X_L - 40)^2}$

$(50)^2 = (30)^2 + (X_L - 10)^2$

$(X_L - 10)^2 = 80 \times 20 : X_L - 10 = 40; X_L = 50\Omega$

Q.17 Answer:

$10^{-8}\mathrm{C}$

Solution:

Put values of ϵ**, we get**

$E = E_0 = \frac{10^{-9}}{10^{-11}} = 100$ **Volt** $\rightarrow$ **(2) Put** E **in**

Reson **ance** $W = \left(\frac{1}{LC}\right)^{y/2} = 10^8\ Q_{\max} = \frac{I_{\max}}{\omega} = 10^{-8}$

Q.18 Answer:

$\sqrt{1000}\ \mathrm{mA}$

Solution:

Charge on the capacitor, $q_0 = CV = 10^{-6}$ **coulomb,**$I_0 = q_0\omega$

Now, $\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10^{-9}}} = \left(10^9\right)^{1/2}$

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$\therefore I_0 = \left(10^9\right)^{1/2} \times \left(1 \times 10^{-6}\right) \text{A} = \sqrt{1000}\,\text{mA}$

Q.19 Answer:

$\left(1 - e^{-1}\right)$

Solution:

$\tau = \frac{L}{R} = 2\,\text{s}, \quad i_0 = \frac{5}{5} = 1\,\text{A}; \quad i = i_0\left(1 - e^{\frac{1}{\tau}}\right) = 1 - e^{-1} \quad (\because t = \tau = 2\,\text{s})$

Q.20 Answer:

100 V

Solution:

$Z = \sqrt{R^2 + X_L - X_C{}^2}$

Here $X_L = 2\pi fL = 2 \times 3.14 \times 500 \times 8.1 \times 10^{-3} = 25.4\ \Omega$

and $X_C = \frac{1}{2\pi fC} = \frac{1}{2 \times 3.14 \times 500 \times 12.5 \times 10^{-6}} = 25.4\ \Omega$

$\therefore\ Z = \sqrt{10^2 + 25.4 - 25.4^2} = 10\ \Omega$

Now $i_{rms} = \frac{E_{rms}}{Z} = \frac{100}{10} = 10\ \text{A}$

$\therefore \quad V_R = i_{rms} \times R = 10 \times 10 = 100\ \text{V}$

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